INJECTIVE QUASI-COHERENT SHEAVES ARE ACYCLIC ETALE-LOCALLY AND HAVE VANISHING HIGHER DIRECT IMAGES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: triage_override: the author's '(details omitted)' sits on 'V x_Y X is qcqs etale over X', justified by etale base-change stability plus quasi-compactness of morphisms from Noetherian spaces; part (1) additionally leans on the delivered L48106 structure lemma

Editorial note. The label lemma-injective-pushforward names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Lemma lemma-injective-colimit-direct-sum-pushforwards-artin (delivered, L48106): every injective in QCoh(O_X) is a direct summand of a colimit of pushforwards from affine closed-ish subschemes over a Noetherian scheme... (the structure lemma used to decompose J)
- Cohomology of Spaces, Lemma spaces-cohomology-lemma-colimits: H^p commutes with filtered colimits on quasi-compact quasi-separated spaces
- Cohomology of Spaces, Lemma spaces-cohomology-lemma-affine-base-change: affine base change identification of the pullback
- Cohomology on Sites, Lemma sites-cohomology-lemma-apply-Leray: Leray spectral sequence

Source reference. Original reference 09TJ: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 2868–2868. Prerequisite pointer invoked at 2932; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0733: sources/stacks-chow/tags.txt, source lines 9047–9047. Prerequisite pointer invoked at 2952; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 07U8: sources/stacks-chow/tags.txt, source lines 9963–9963. Prerequisite pointer invoked at 2947; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 073E: sources/stacks-chow/tags.txt, source lines 9058–9058. Prerequisite pointer invoked at 2937; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 09TF: sources/stacks-chow/tags.txt, source lines 12388–12388. Prerequisite pointer invoked at 2939; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let X be a Noetherian algebraic space over S. If $\mathcal{J}$ is an injective object of $QCoh(\mathcal{O}_X)$, then*

1. *$H^p(U, \mathcal{J}|_U) = 0$ for $p > 0$ and for every quasi-compact and quasi-separated algebraic space U étale over X,*
2. *for any morphism $f : X \to Y$ of algebraic spaces over S with Y quasi-separated we have $R^p f_* \mathcal{J} = 0$ for $p > 0$.*

Proof. Proof of (1). Write $\mathcal{J}$ as a direct summand of $\operatorname{colim} \mathcal{F}_i$ with $\mathcal{F}_i = (Z_i \to X)_* \mathcal{G}_i$ as in Lemma 09TJ. It is clear that it suffices to prove the vanishing for $\operatorname{colim} \mathcal{F}_i$. Since pullback commutes with colimits and since U is quasi-compact and quasi-separated, it suffices to prove $H^p(U, \mathcal{F}_i|_U) = 0$ for $p > 0$, see Cohomology of Spaces, Lemma 073E. Observe that $Z_i \to X$ is an affine morphism, see Morphisms of Spaces, Lemma 09TF. Thus

$$\mathcal{F}_i|_U = (Z_i \times_X U \to U)_* \mathcal{G}_i' = R(Z_i \times_X U \to U)_* \mathcal{G}_i'$$

where $\mathcal{G}_i'$ is the pullback of $\mathcal{G}_i$ to $Z_i \times_X U$, see Cohomology of Spaces, Lemma 07U8. Since $Z_i \times_X U$ is affine we conclude that $\mathcal{G}_i'$ has no higher cohomology on $Z_i \times_X U$. By the Leray spectral sequence we conclude the same thing is true for $\mathcal{F}_i|_U$ (Cohomology on Sites, Lemma 0733).

Proof of (2). Let $f : X \to Y$ be a morphism of algebraic spaces over S. Let $V \to Y$ be an étale morphism with V affine. Then $V \times_Y X \to X$ is an étale morphism and $V \times_Y X$ is a quasi-compact and quasi-separated algebraic space étale over X (details omitted). Hence $H^p(V \times_Y X, \mathcal{J})$ is zero by part (1). Since $R^p f_* \mathcal{J}$ is the sheaf associated to the presheaf $V \mapsto H^p(V \times_Y X, \mathcal{J})$ the result is proved. □